



# Marine Biology

# 海洋生物学



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## • Major Phyla of Marine Invertebrates

- |                       |                            |
|-----------------------|----------------------------|
| 1. Sponges (海绵动物)     | 10. Molluscs (软体动物)        |
| 2. Cnidarians (腔肠动物)  | 11. Bryozoans (苔藓虫)        |
| 3. Flatworms (扁形虫)    | 12. Phoronids (帚虫)         |
| 4. Ribbon worms (纽虫)  | 13. Brachiopods (腕足类)      |
| 5. Nematodes (线虫)     | 14. Arrow worms (箭虫)       |
| 6. Annelids (环节动物)    | 15. Echinoderms (棘皮动物)     |
| 7. Sipunculans (星虫动物) | 16. Hemichordates (半索海生动物) |
| 8. Echiurans (螠虫动物)   | 17. Tunicates (被囊动物)       |
| 9. Arthropods (节肢动物)  | 18. Lancelets (文昌鱼)        |

## 11.4 Cephalopods

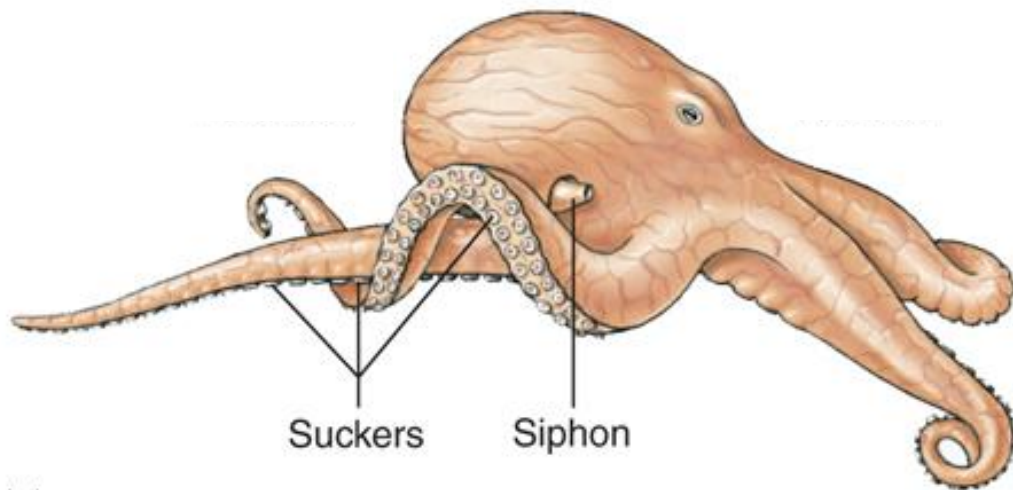
- All 650 species marine, including Squid, Octopus, Cuttlefish & Nautilus (鹦鹉螺)



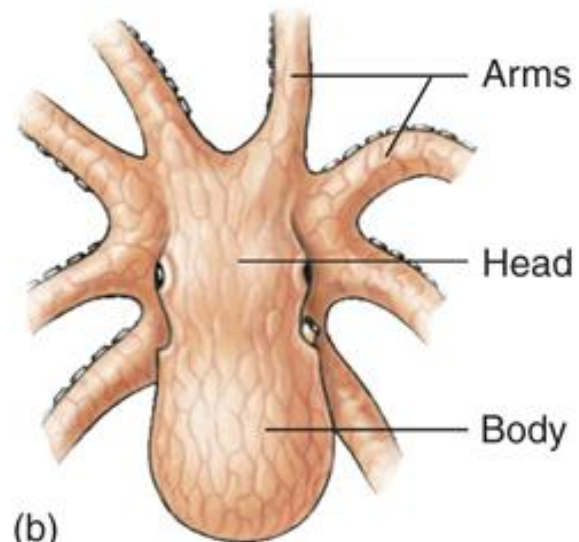
- **Most advanced invertebrates**
- **Shell internal or absent** in most, **thick mantle** covers the body, well developed eyes, adapted tentacles
- Use **beak-like jaws** and **radula** to crush or rip prey
- **Fast swimming predators** due to water jet propulsion



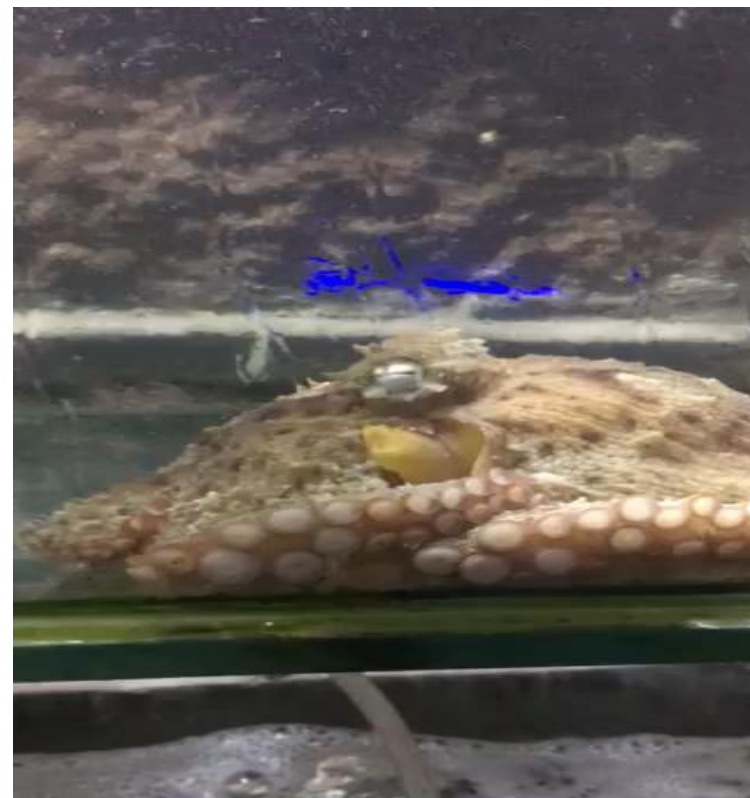
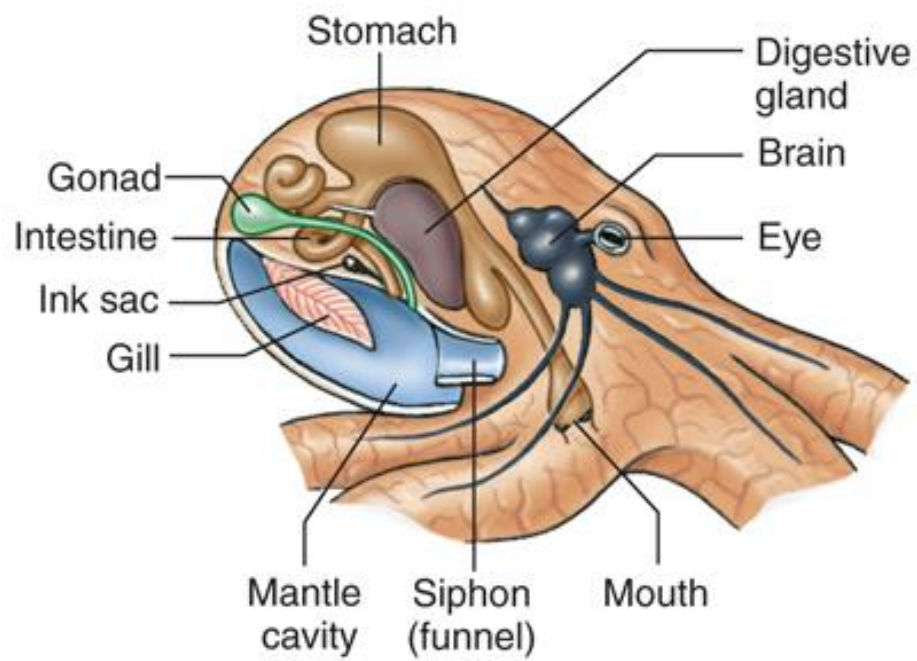


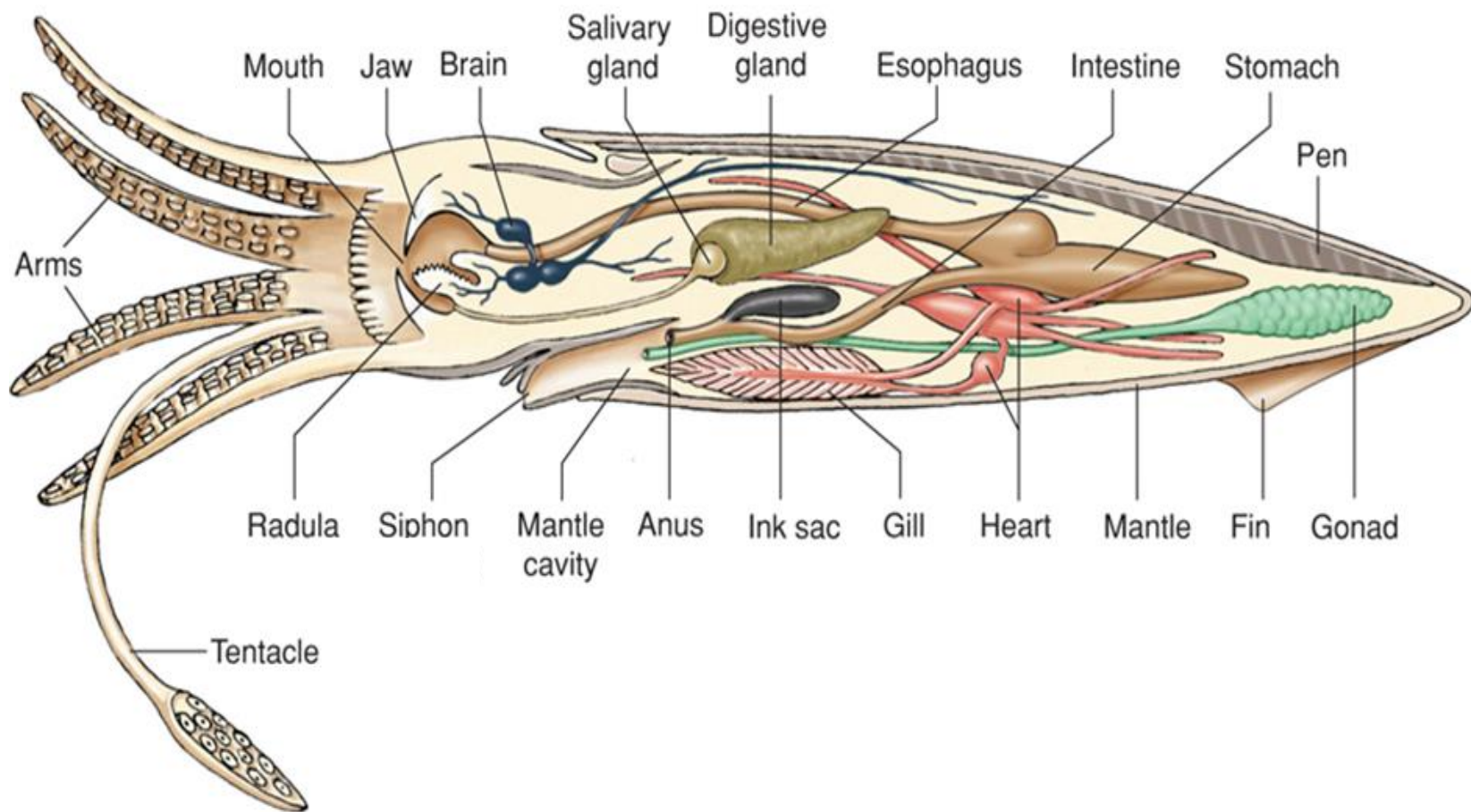


(a)



(b)





**Squid**



**Cuttlefish**

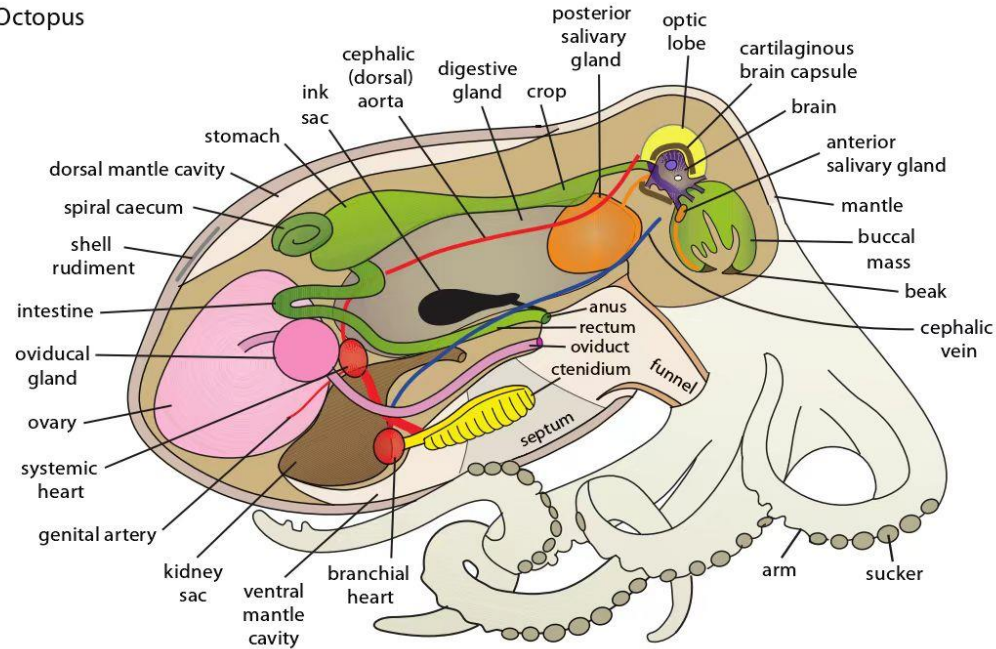


**Octopus**

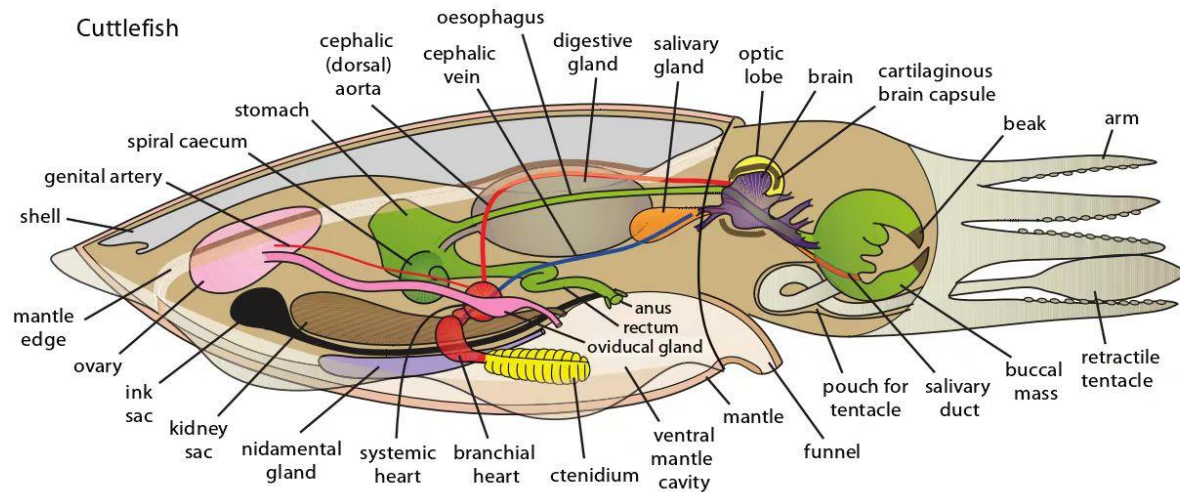




## Octopus



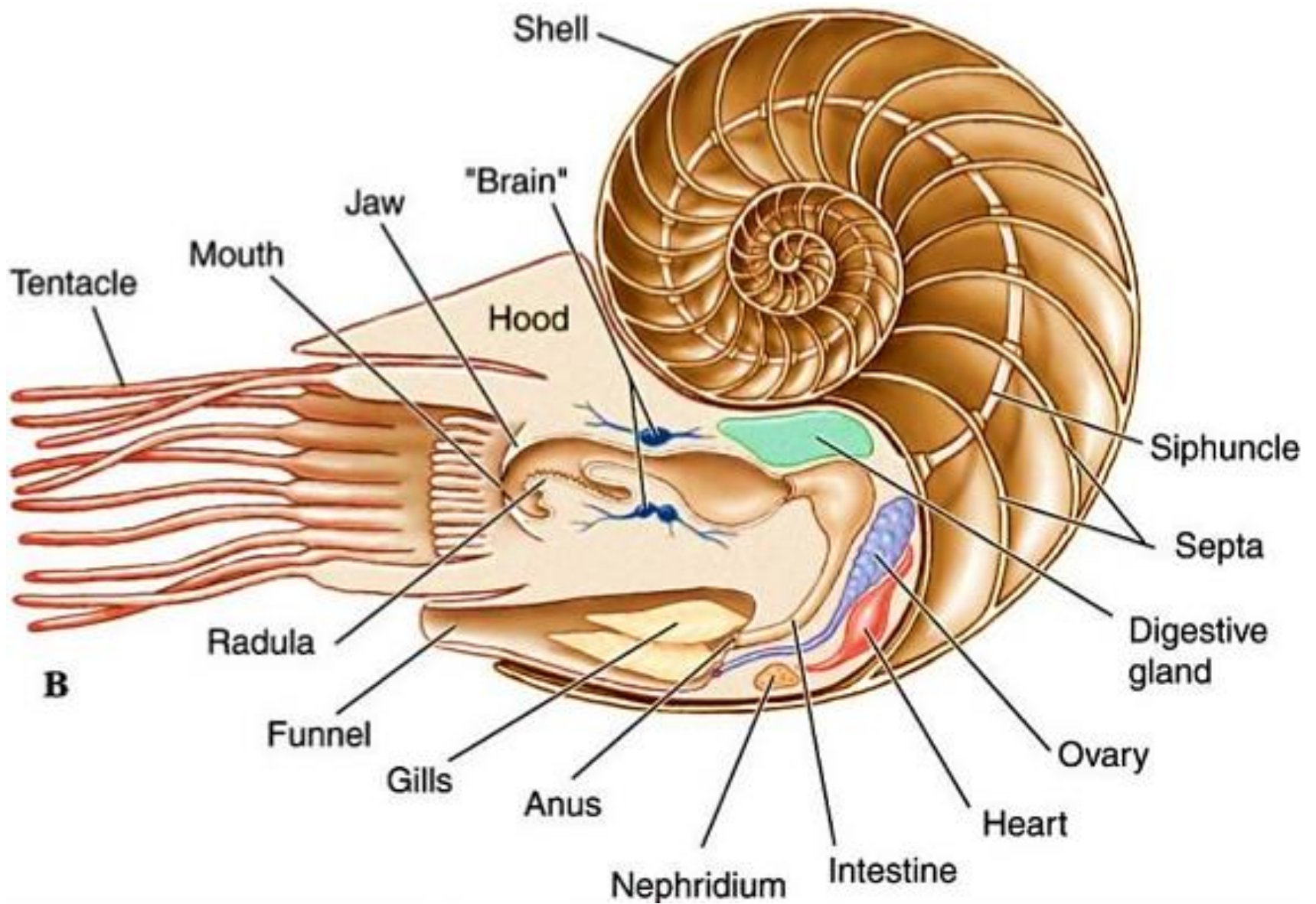
## Cuttlefish

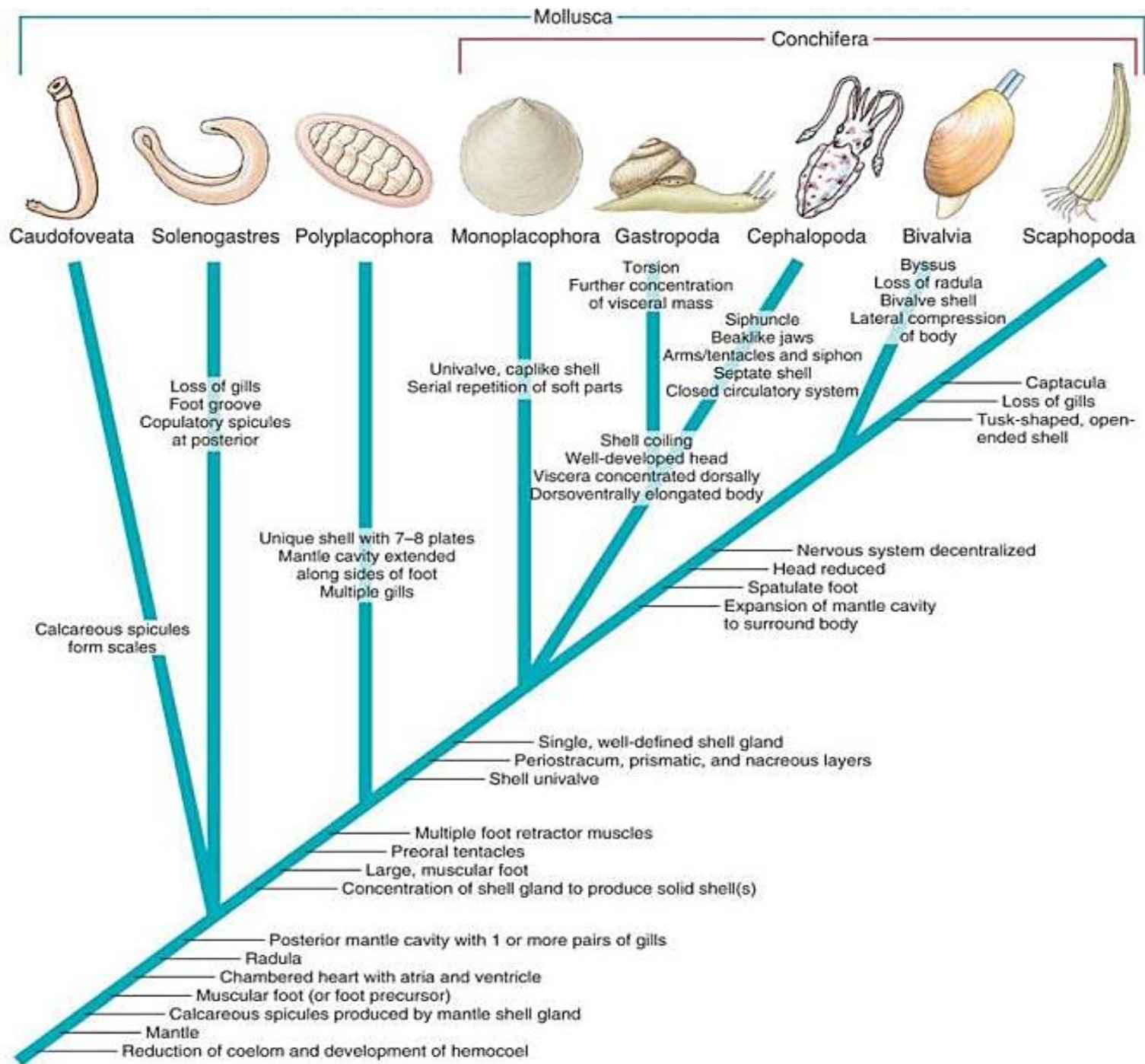


**Nautilus:** survived relatively unchanged for hundreds of millions of years (living fossils), benthic animals (2 families, 6 species)









## 11.5 Biology of Mollusks

- **Feet of Mollusks:** muscular organ, shaped and used differently by different species
  - 1) A **clam** uses its hatchet-shaped foot for digging
  - 2) A gastropod has a flat foot it uses for crawling





### 3) **Octopus** Foot (Arm/Tentacles):

The foot in octopus and squids is modified into many tentacles that are attached to the animal's head. Octopus and squids use their tentacles for moving and for grasping and holding the prey, as well as mating

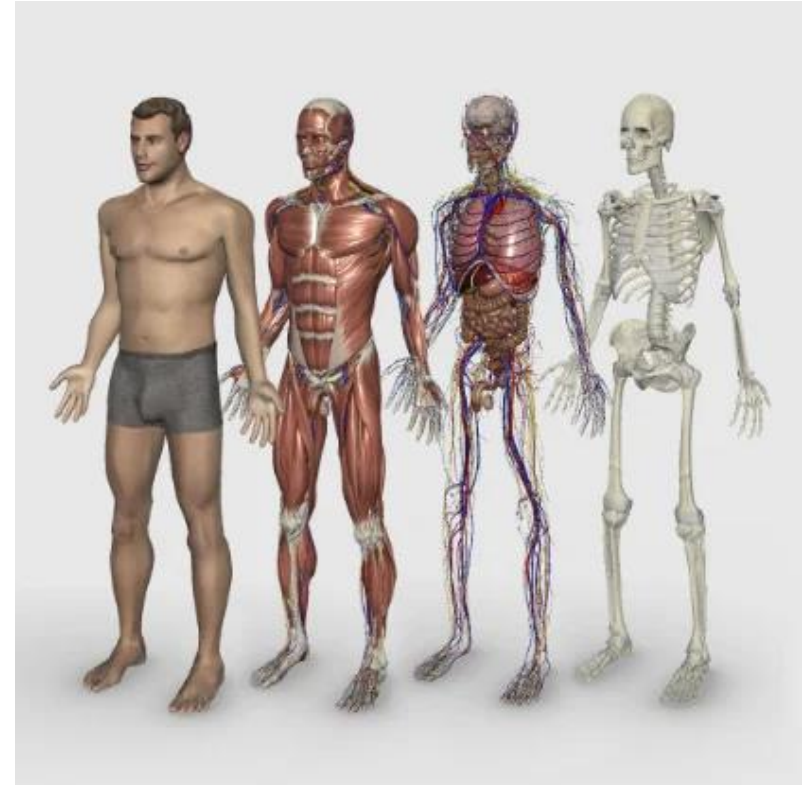
### 4) **Nautilus** foot: have many cirri (60-90), varying with a species, lack pads or suckers, prey by their ridged surface



- **Mantle and Shell of Mollusks**

- ❑ The area of the mollusk called the “mantle” makes the shell for the animal (if it has a shell)
- ❑ Mantle creates the colors and patterns on the shells
- ❑ The shell is an exoskeleton (外骨骼), even though it is completely surrounded by soft tissue in some mollusks
- ❑ Because the shell is continually produced, it grows with the animal

# What is the difference between an exoskeleton and an endoskeleton



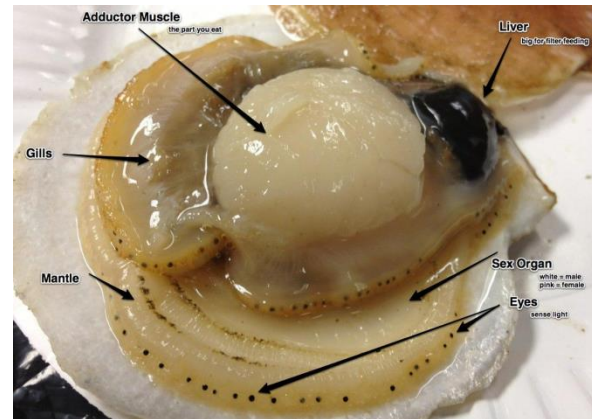


## ❑ Chiton's Mantle:

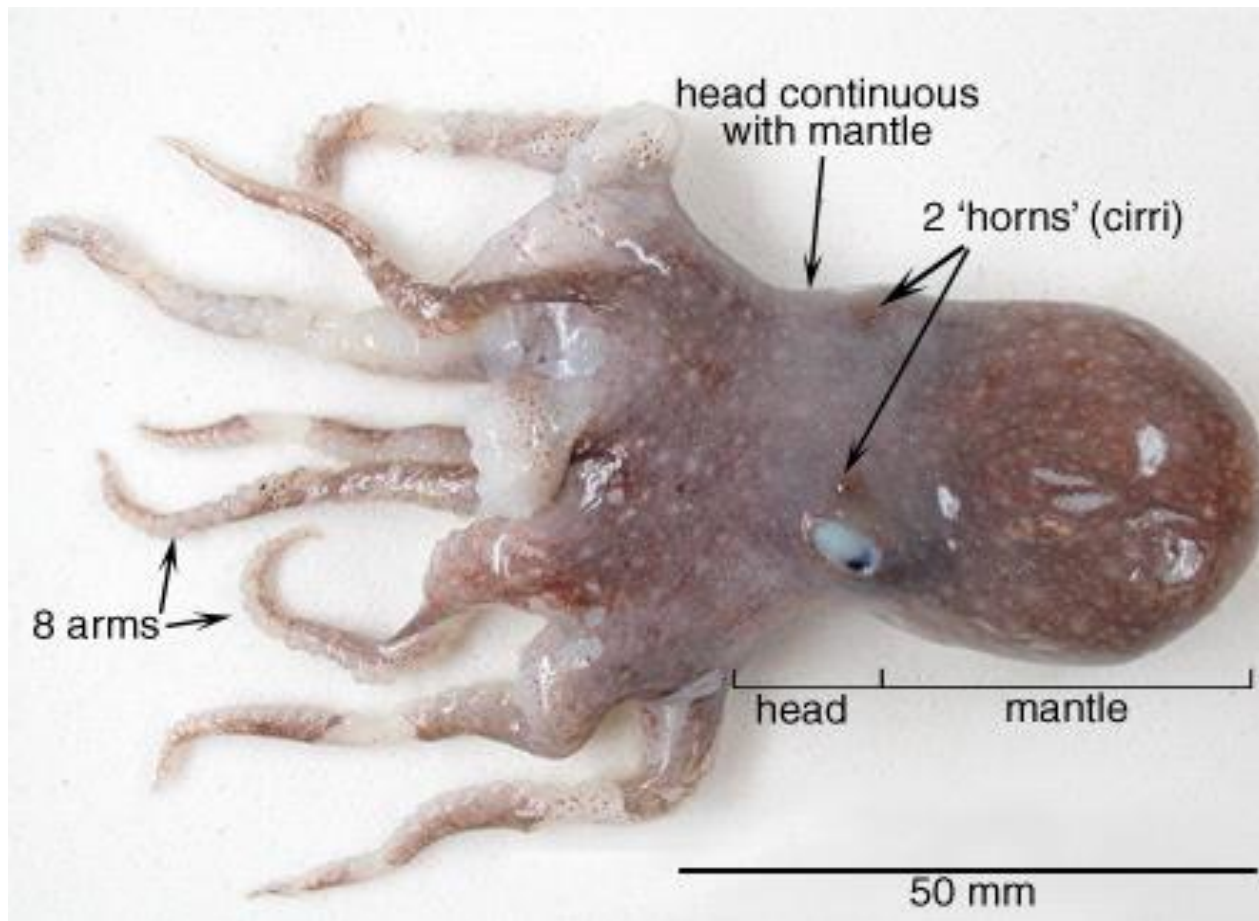
Produces about 7-8 separate plates that cover the body. Have joints between each of these plates to allow it to curl up in a ball



- ## ❑ Mantle in bivalves and gastropods cannot be seen, since it is the inner layer of the shell.
- Bivalves make two distinct shells, gastropods produce a single shell in a spiral shape



- Cephalopod also have a mantle. But the mantle is opposite and covers the shell instead of being inside the shell. This shell is inside their bodies and is called a pen



- **Feeding and Digestion**

- 1) Separate mouth and anus
- 2) Digestion involves salivary (唾液) and digestive glands that release digestive enzymes for metabolisms
- 3) Digestion is partly extracellular in gut cavity and partly intracellular in the digestive glands
- 4) Some gastropods that feed on seaweeds can keep the chloroplasts in digestive gland and use it for gastropod's nourishment (营养物) via photosynthesis
- 5) Grazers can efficiently process large amounts of hard-to-digest plant material
- 6) Carnivorous snails: use its radula to prey and their digestion is extracellular and takes in the stomach



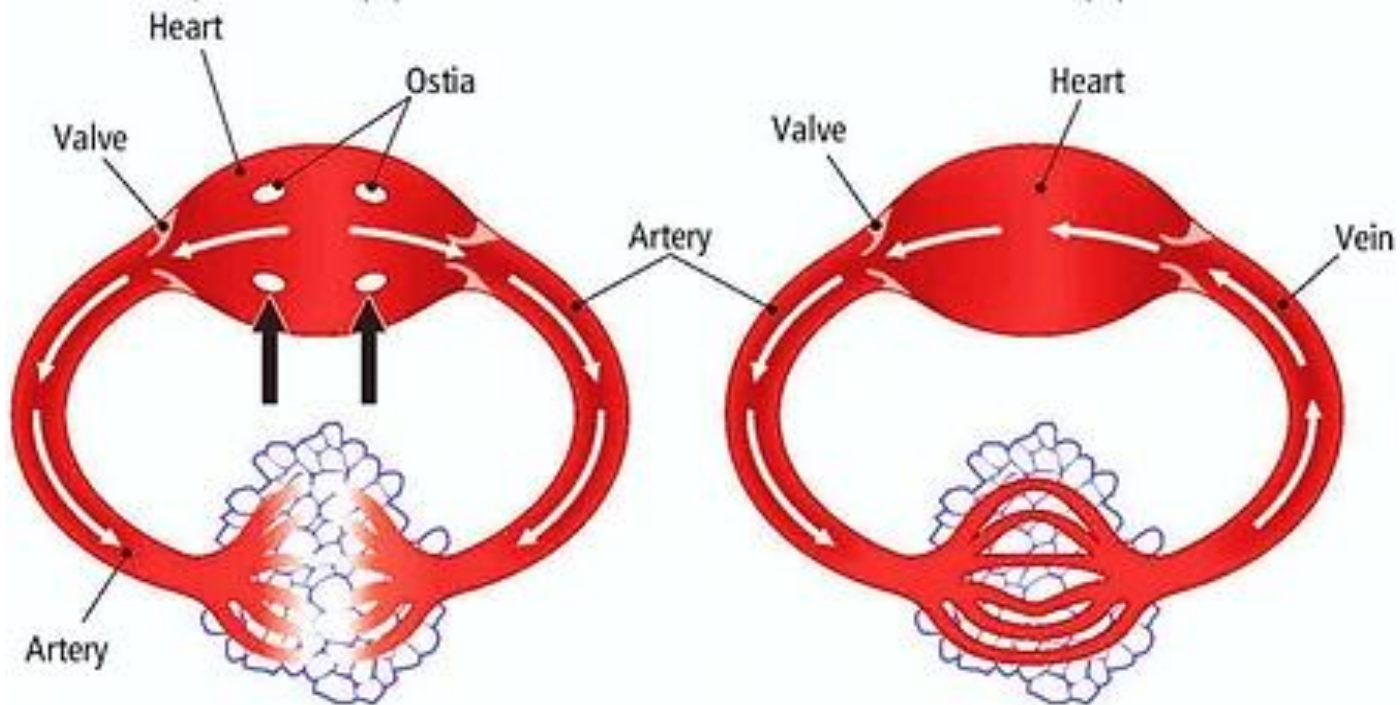
## 7) Bivalves:

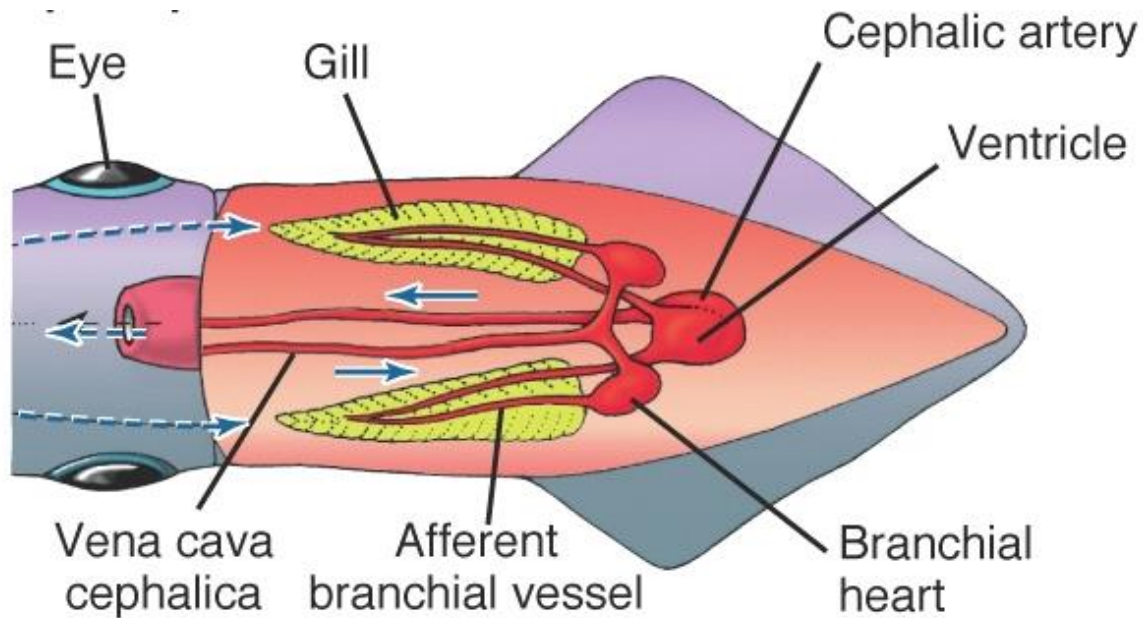
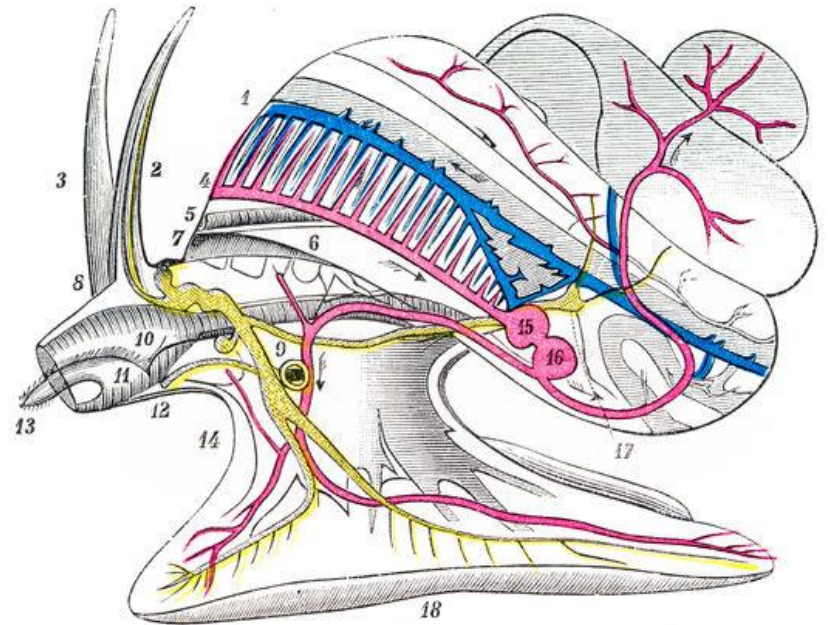
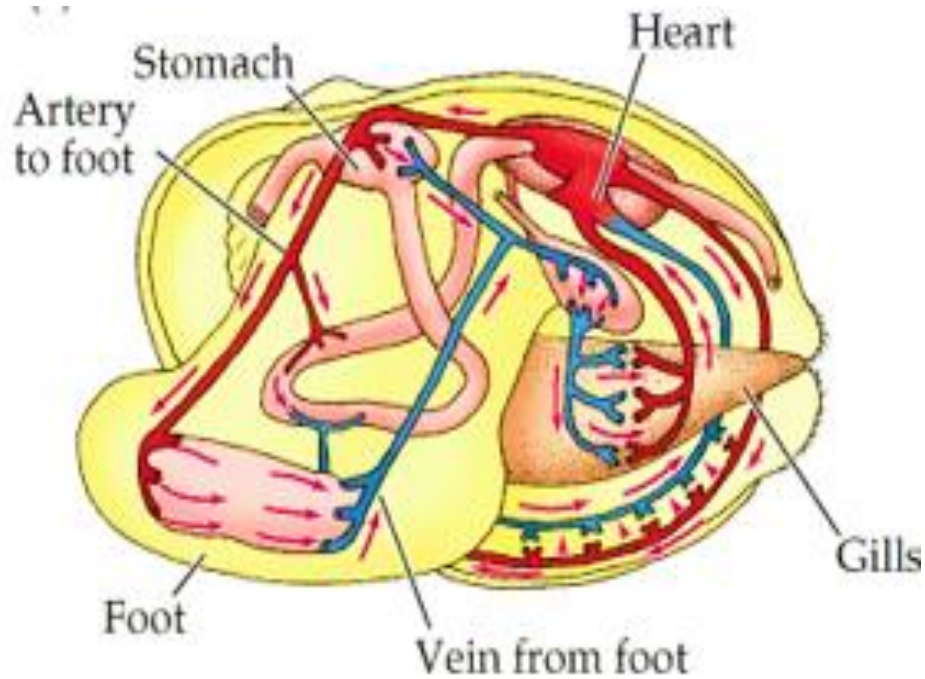
- ❑ Using the cilia on the gills to filter and sort out food particles
- ❑ No radula
- ❑ Food enters the mouth trapped in long strings of mucus
- ❑ Crystalline style (enzyme-secreting rod in the stomach) continually rotates the food to help in its digestion
- ❑ The stomach contents eventually pass into a large digestive gland for intracellular digestion

## 8) Cephalopods:

- ❑ Carnivores
- ❑ Stomach is sometimes connected to a sac in which digestion is rapidly and efficiently completed
- ❑ **It is entirely extracellular**

- **Circulatory system:** To transport nutrients and oxygen; muscular heart (dorsal) pumps blood to all tissues
- ❑ Most mollusks have an **open circulatory system** in which blood flows out of vessels into open blood spaces
- ❑ **Cephalopods have close circulatory system** in which the blood always remains in vessels and can be more effectively directed to oxygen-demanding organs such as the brain

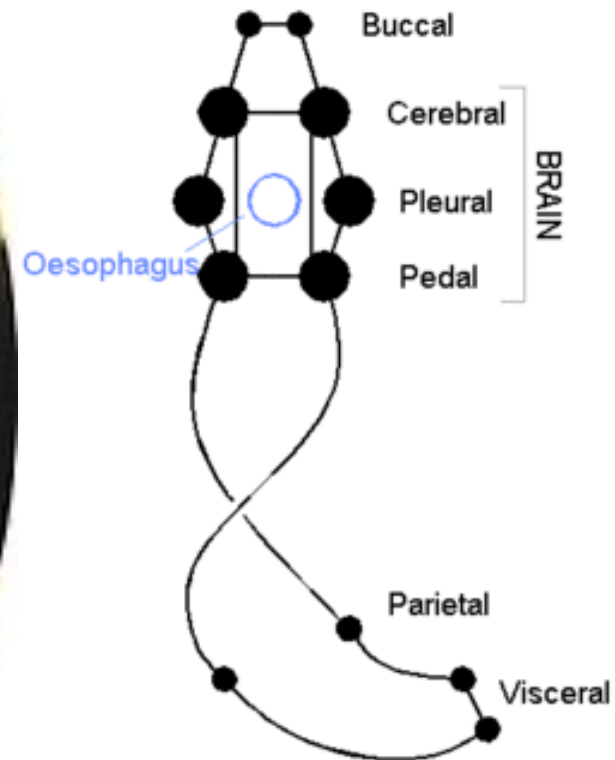
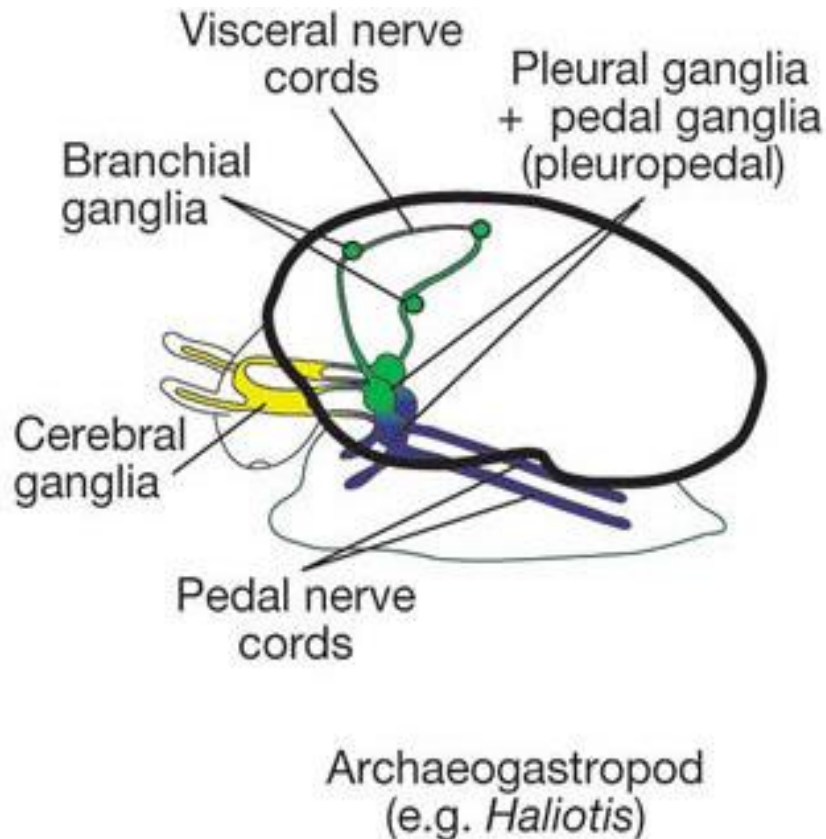




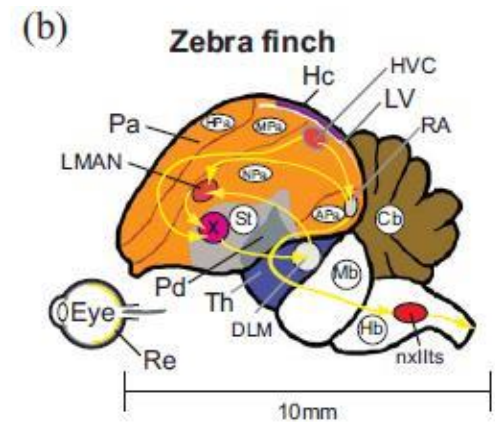
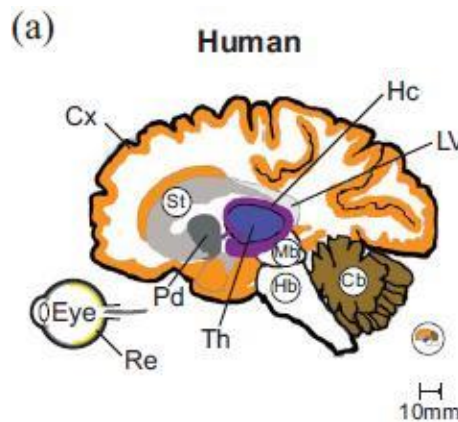
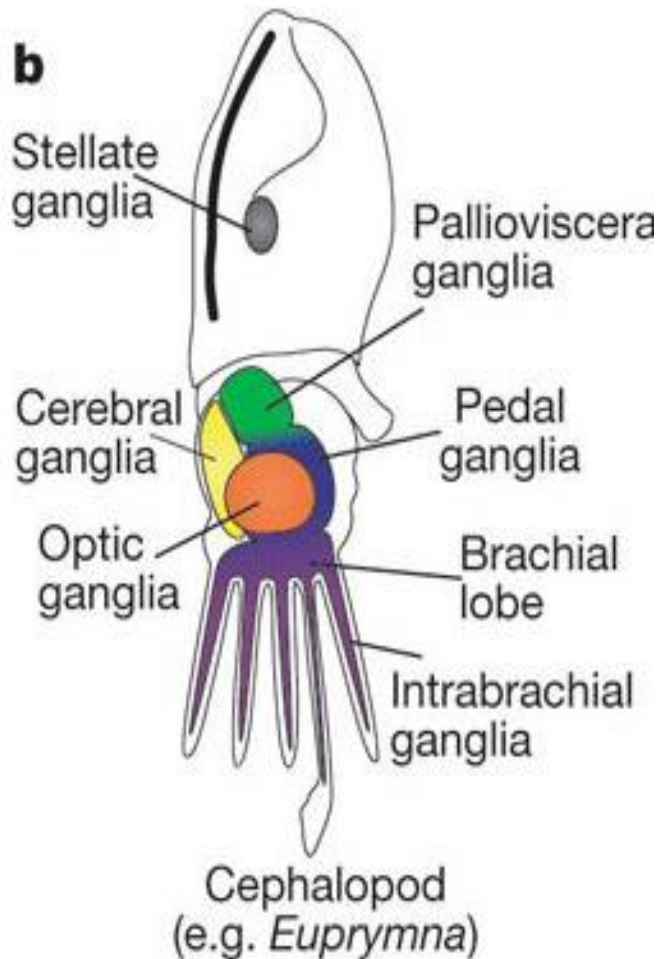


- Nervous System and Behavior

- 1) Gastropods and bivalves do not have a single brain but several sets of ganglia (神经节), or 'local brains' clusters of nerve cells located in different parts of the body



- 2) Cephalopods have the most complex nervous system not just of mollusks but of all invertebrates. Separate local brains are fused into a single, large brain that coordinated and stores information received from the environment



- Paul the Octopus-the psychic for World Cup 2010





3) Different behaviors can be controlled by particular regions of the brain (as in human):

Example: Display color changes correlated with particular behaviors and moods, from intricate sexual displays to camouflage (伪装) to prey



## • Reproduction and Life History

- 1) Have separate sexes, but some species are hermaphrodites
- 2) In **bivalves, chitons and some gastropods**, sperm and eggs are released into the water and fertilization is external
- 3) In **cephalopods and most gastropods**, the fertilization is internal: male uses a **modified arm** to transfer a spermatophore (an elongated packet of sperm, 精囊) to the female
- 4) In many cephalopods, part or all of development takes place within strings or capsules of eggs
- 5) Female octopuses protect their eggs until they hatch



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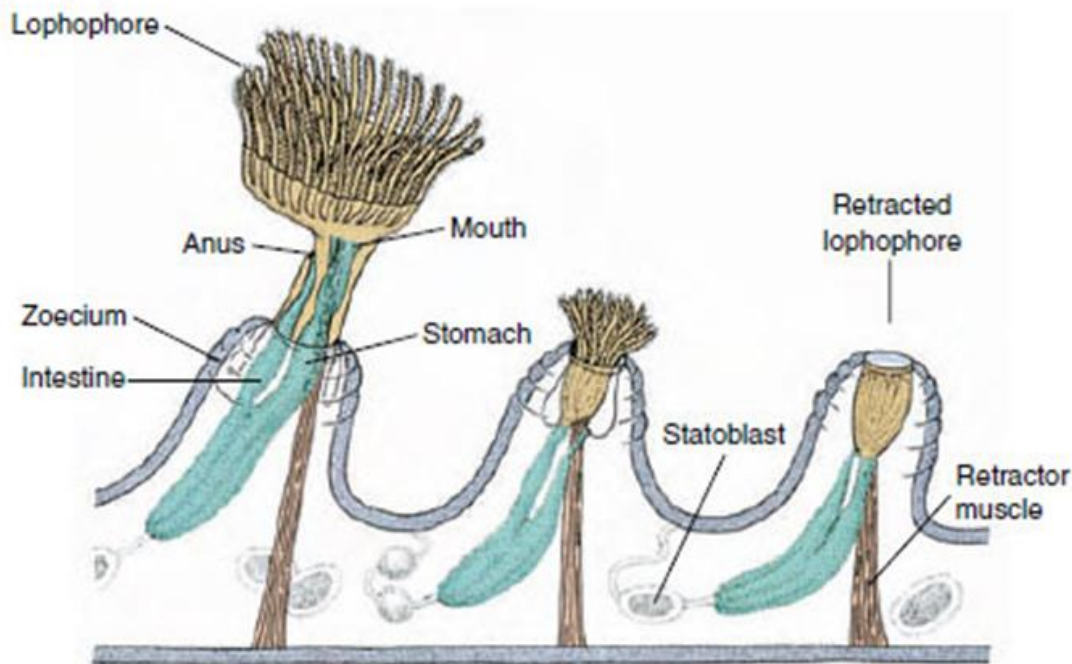
## 12: Lophophore (触手冠动物)

- Bryozoans (plus phoronids and brachiopods) possess a unique **feeding structure**——lophophore
- The **lophophore** (触手冠) is an extension of the body and ends in a tentacle structure that surrounds the mouth, function in both feeding and respiration

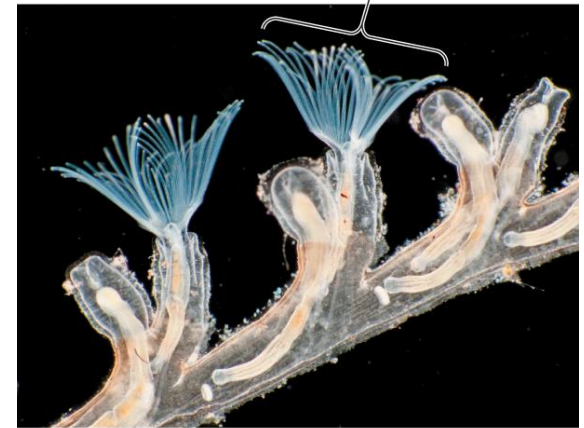


## 12.1 The Bryozoans (苔蘚虫)

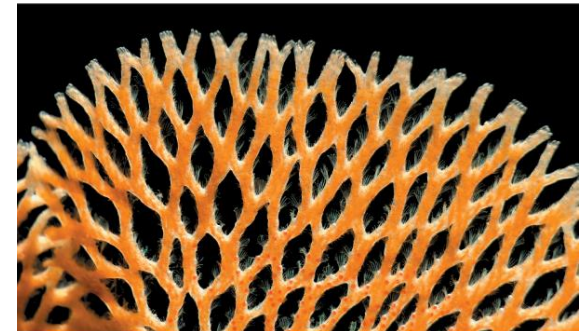
- 4500 species, **all marine**, also called moss animals
- Small, colonial animals
- Some are encrusting forms, others form a branching network that looks like a small tree
- Suspension feeders



(A) *Plumatella repens* Lophophore composed of tentacles



(B) *Reteporella couchii*

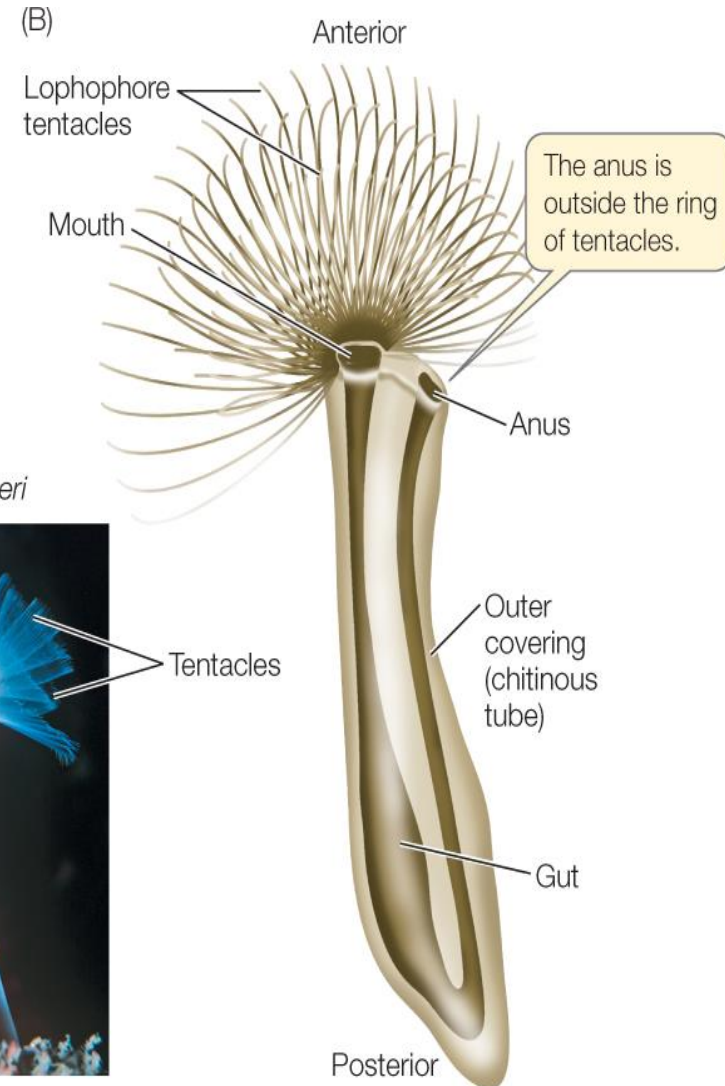


## 12.2 The Phoronida (帚虫)

- 20 species
- Found in shallow, coastal waters
- Phoronids make tubes where they live
- Somewhat resemble Annelid worms externally
- Lophophore present



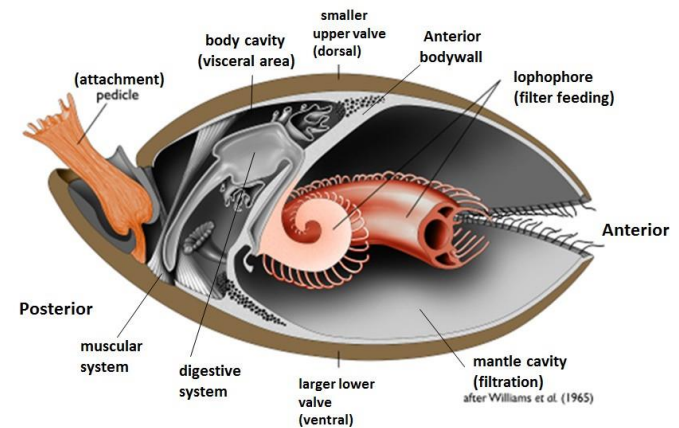
(A) *Phoronis mulleri*





## 12.3 The Lamp Shell /Brachiopods (腕足类)

- 300 species, **exclusively marine**
- Many live in deep sea or at polar regions
- Two shells or “valves” enclose the body
- Possess lophophore
- Well known from fossil record, indicates there were many more species in geologic past



# 13: The Arrow Worms (Chaetognatha, 毛颚动物门, 箭虫)

- 100 species, **all marine**
- These organisms are planktonic, but slightly larger than most plankton
- They are voracious predators that are widely distributed in the marine community
- They will swim in rapid, darting movements to capture prey



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